MAI643: Artificial Intelligence in Medicine
Tutorial 1: Anaconda Platform

University of Cyprus
Department of
Computer Science

Pavlos Antoniou

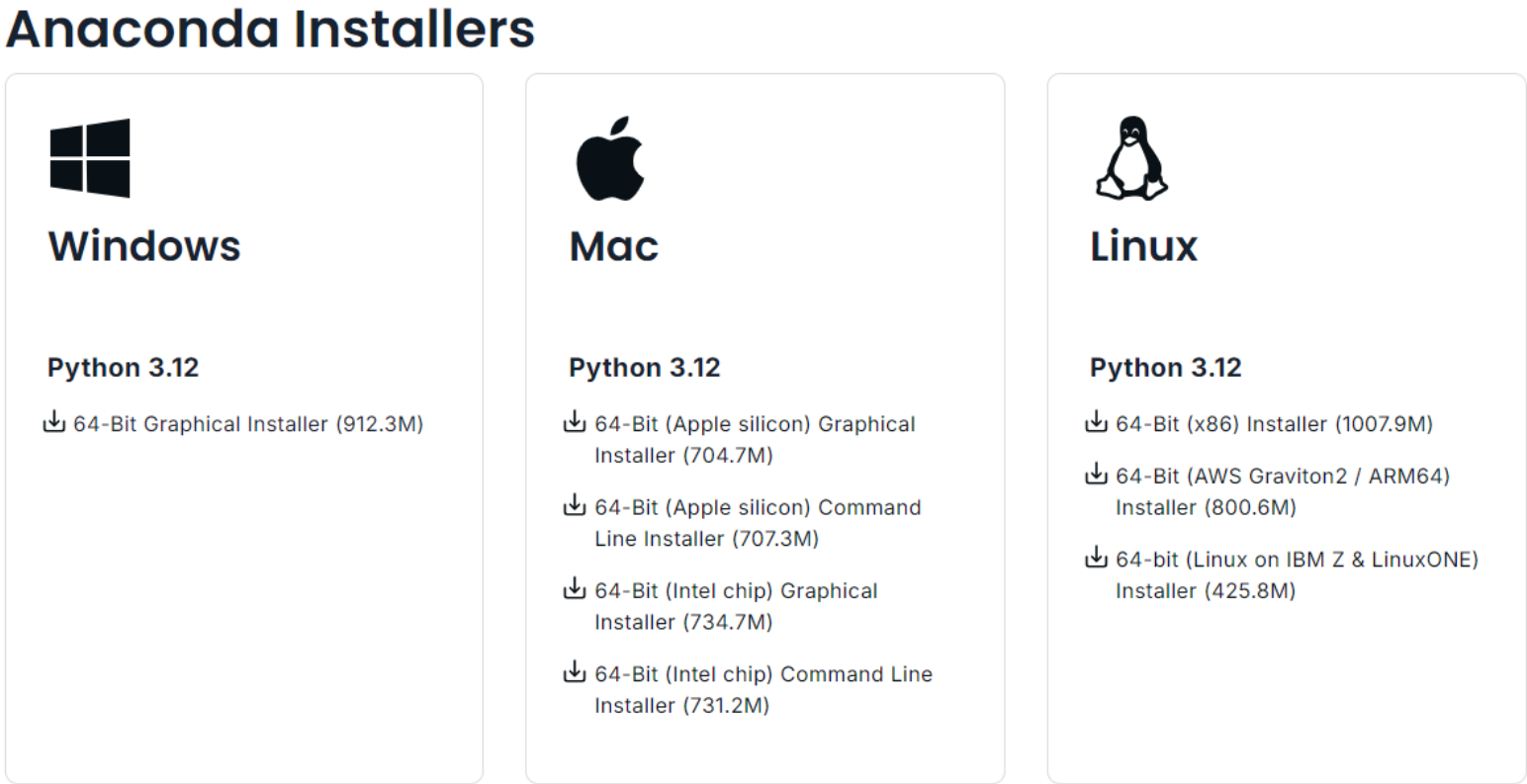
Office: B109, FST01

Available Python Distributions

- Official Python website: https://www.python.org/
  - Executables for Windows, macOS, Unix: https://www.python.org/downloads/
  - Includes basic data types and associated functions to operate on them
  - However, … you need to install numerous Data Science packages (numpy, pandas, sklearn) using *pip* (package management system)
- Anaconda Data Science Platform: https://www.anaconda.com/
  - Free for individuals, open-source, easy-to-install Python distribution with over a 1500 packages (including the package management system *conda*) and a GUI named Anaconda Navigator
    - Through Anaconda Navigator you can gain access to some very useful applications such as Jupyter Notebook and Spyder IDE
  - Powered by Anaconda company, but with free community support

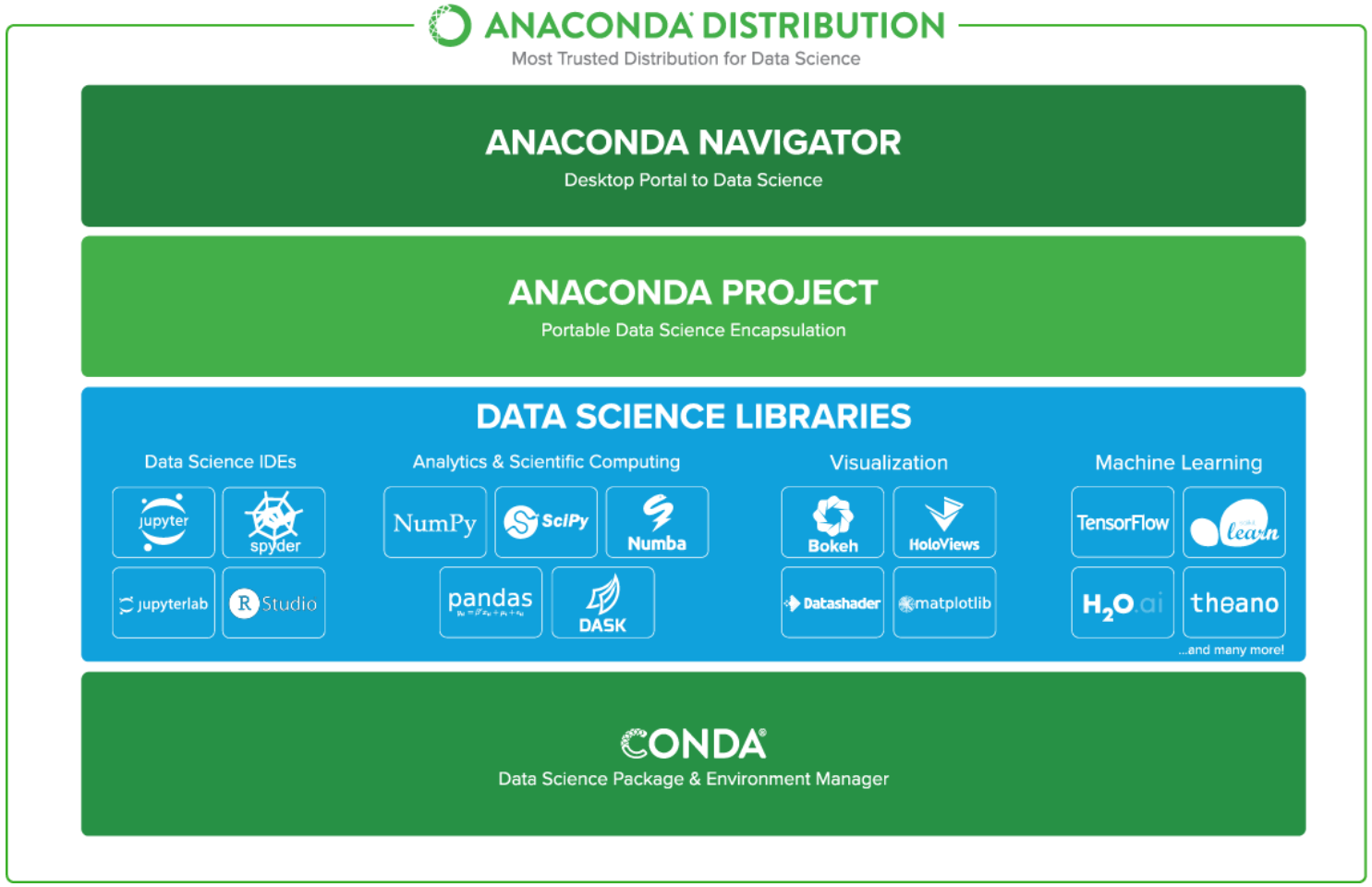
Install Anaconda

- Step 1: Go to https://www.anaconda.com
- Step 2: Download and install the version of Anaconda targeting your Operating System (prefer 64 bit). For all available installers see here: https://www.anaconda.com/download/success



Getting started with Anaconda

- Anaconda Distribution contains conda and Anaconda Navigator as well as Python and hundreds of scientific packages.



Courtesy of https://medium.com/@kumarankita764/new-features-of-anaconda-5-3-5bfdfe9b4240

Getting started with Anaconda

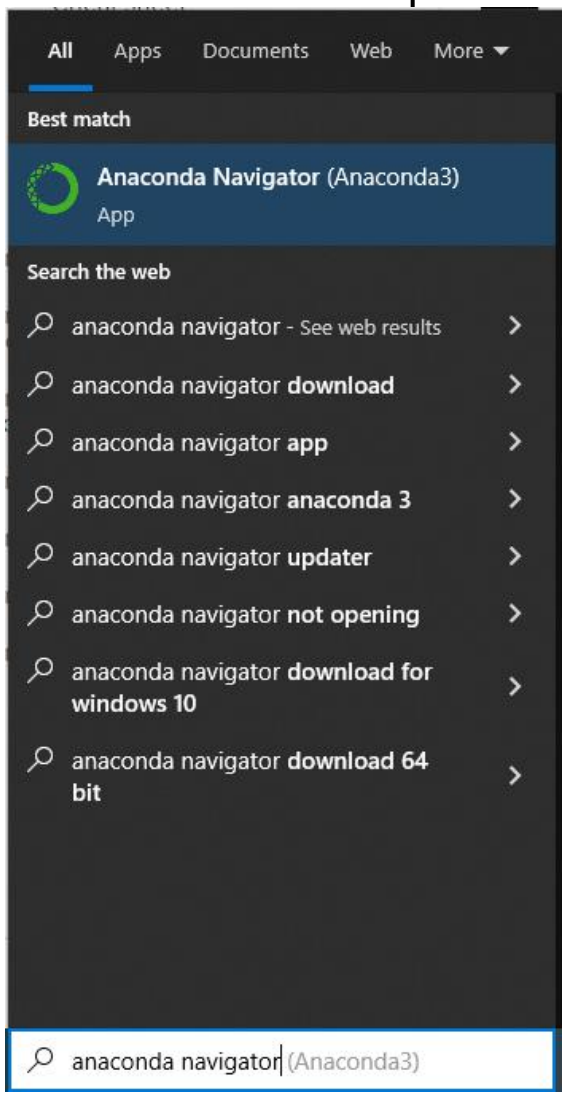
- Conda is as a command line interface **called Anaconda Prompt** on Windows and via terminal on macOS and Linux.
  - Package, dependency and environment management system
    - quickly installs and updates packages and their dependencies

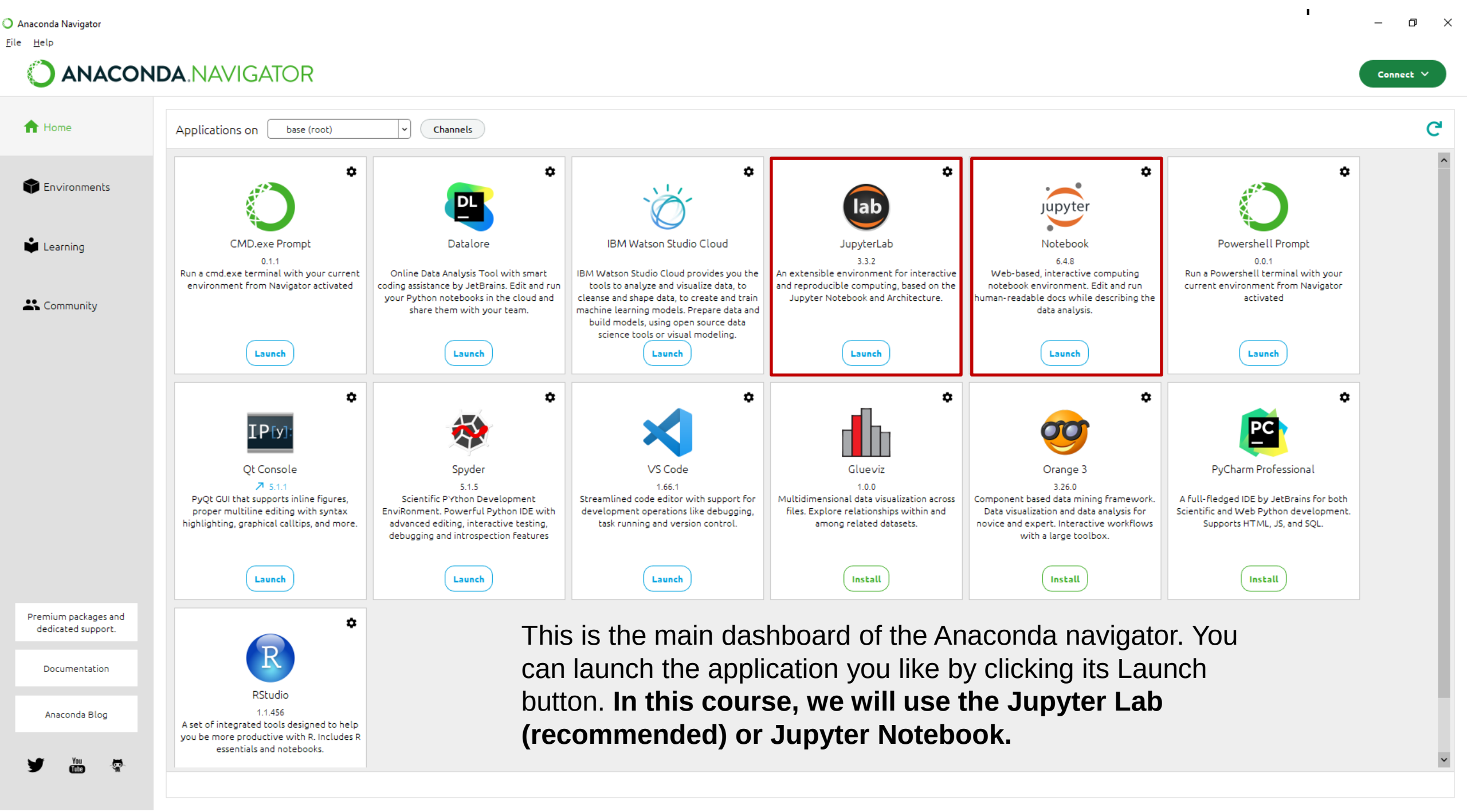
- Navigator is a desktop graphical user interface (GUI) that allows you to launch applications and easily manage conda packages.

Python program using Anaconda Navigator

- Launch Anaconda Navigator on Windows:
  - Go to Start Menu and type “anaconda navigator” →
- Launch Anaconda Navigator on macOS:
  - Open Launchpad, then click the Anaconda Navigator icon:

- Launch Anaconda Navigator on Unix/Linux:
  - Open a terminal and type “anaconda-navigator”

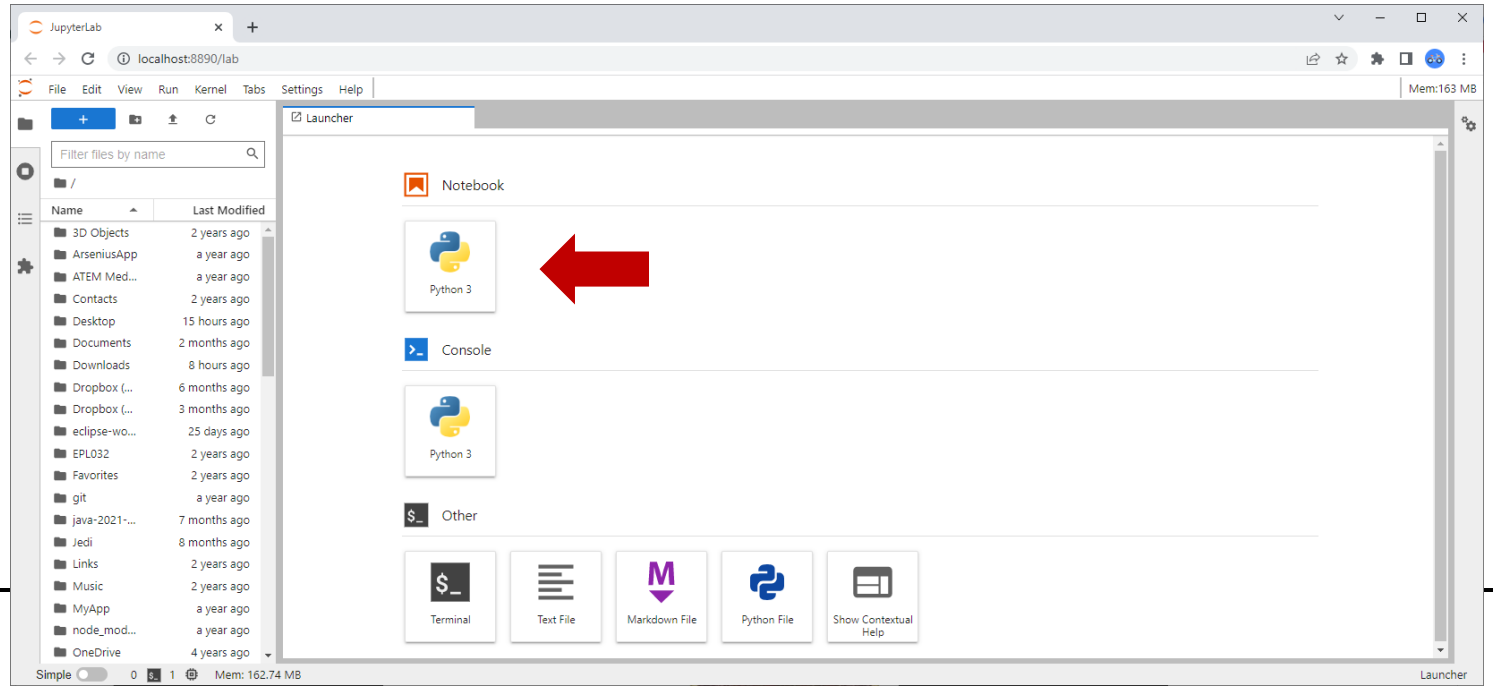




This is the main dashboard of the Anaconda navigator. You can launch the application you like by clicking its Launch button. **In this course, we will use the Jupyter Lab (recommended) or Jupyter Notebook.**

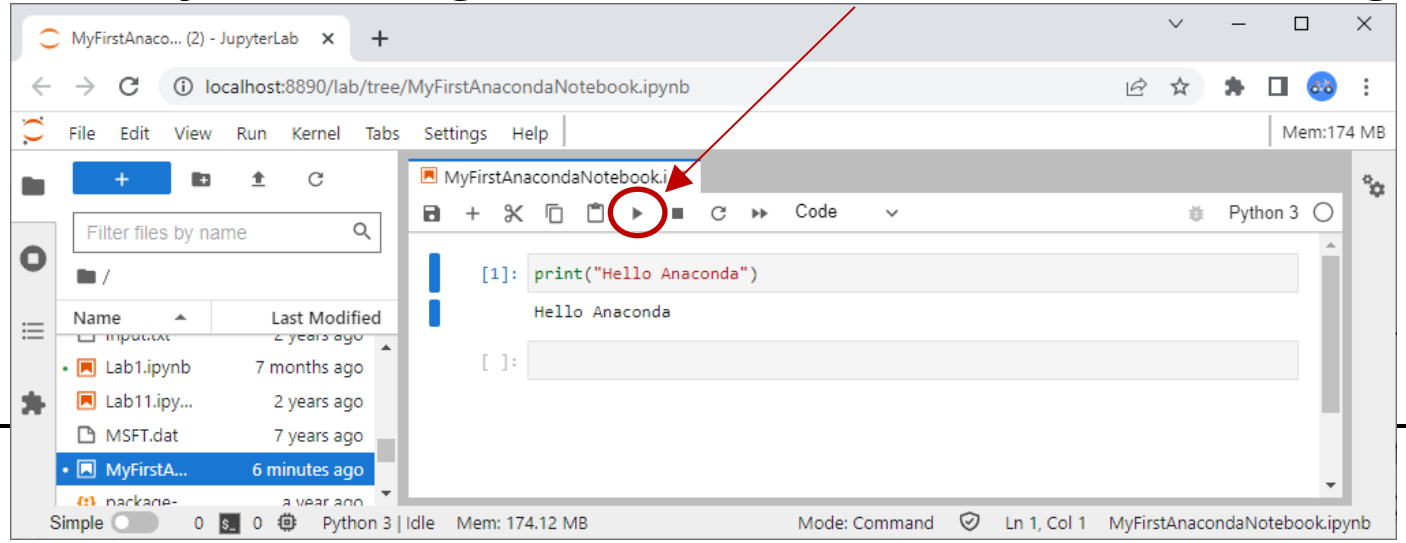
Write/Run Python programs in JupyterLab

- Launch JupyterLab by clicking JupyterLab's Launch button.
- This will launch a new browser window (or a new tab on an existing browser window) showing the Notebook Dashboard.
- In the middle of the Launcher tab there is a button labeled with the Python version you installed. Click it and create a new Notebook.



Run Python in JupyterLab

- Rename your Notebook
  - Right click on the filename (shown on the tab). You can rename it to whatever you'd like, but for this example we'll use MyFirstAnacondaNotebook.ipynb
- In the first line of the Notebook, type `print("Hello Anaconda")`
- Save your Notebook by either clicking the save and create checkpoint icon or by selecting File → Save Notebook in top menu.
- Run your new program by clicking the Run button or selecting Run from the top menu.

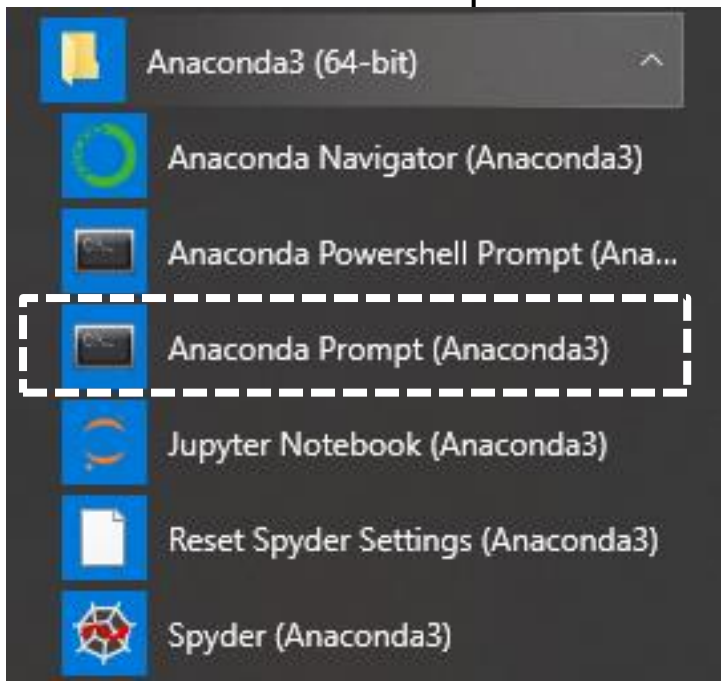


Launch JupyterLab without Navigator

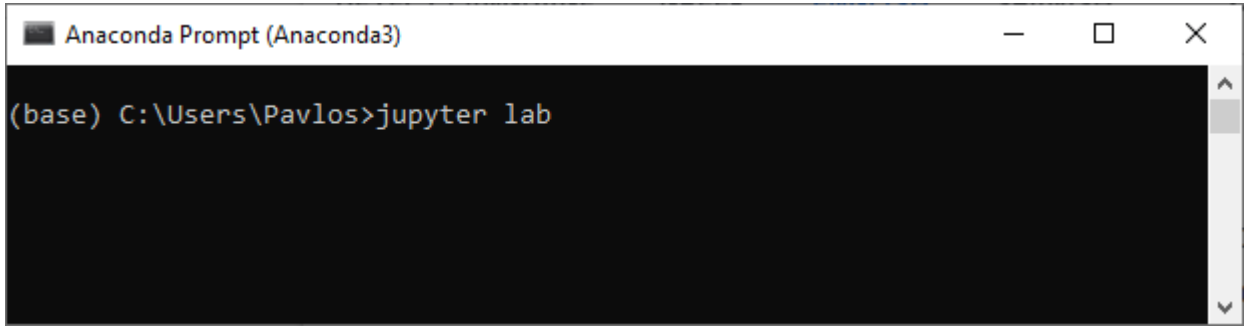
- You can launch JupyterLab without the graphical application Anaconda Navigator which is a heavy (in terms of resources, i.e. RAM and CPU) environment, by using the Anaconda Prompt in Windows or terminal in Linux and MacOS.
- In Windows, Anaconda prompt can be launched as shown in the next slide from the Windows Start Menu.

Launch JupyterLab from the command line

- Launch Anaconda Prompt in Windows (or terminal in Linux / macOS)



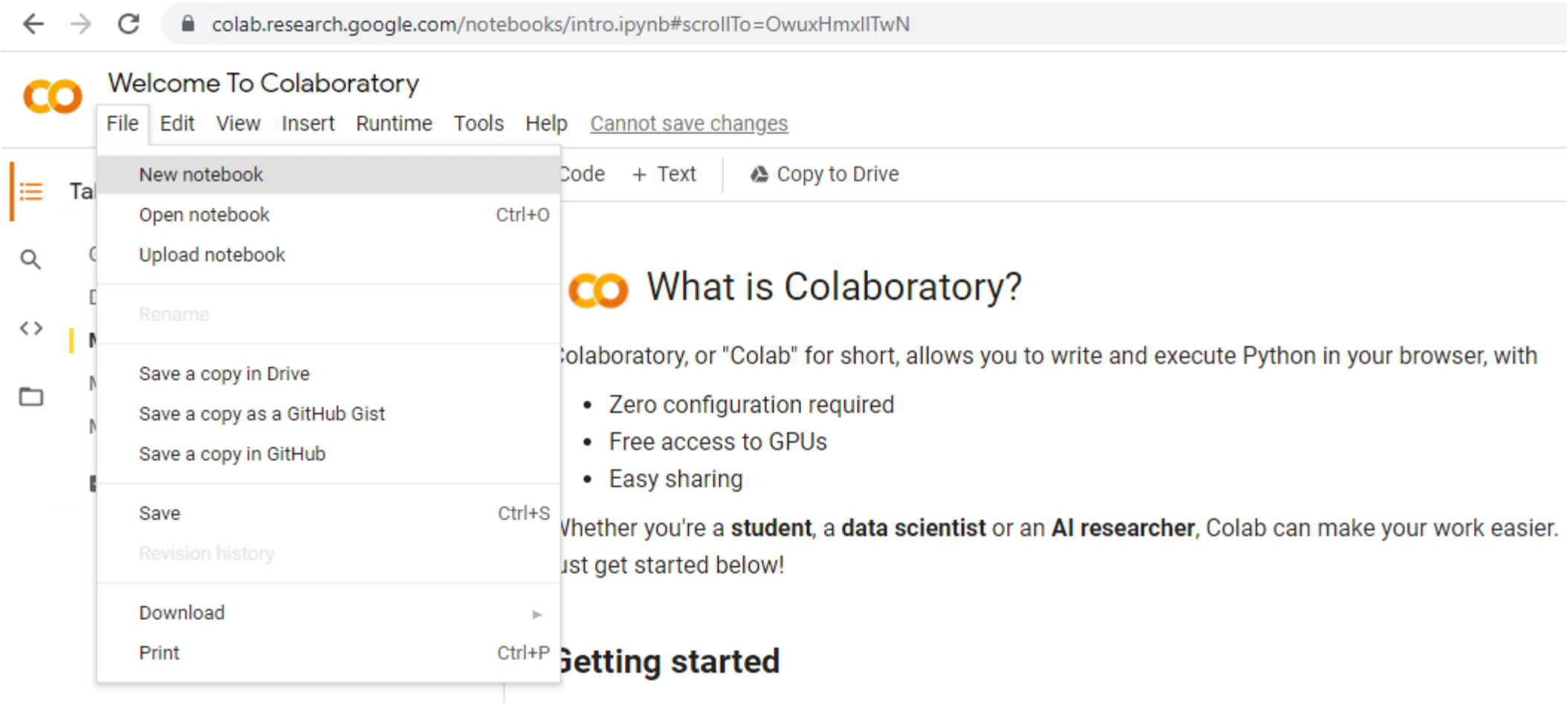
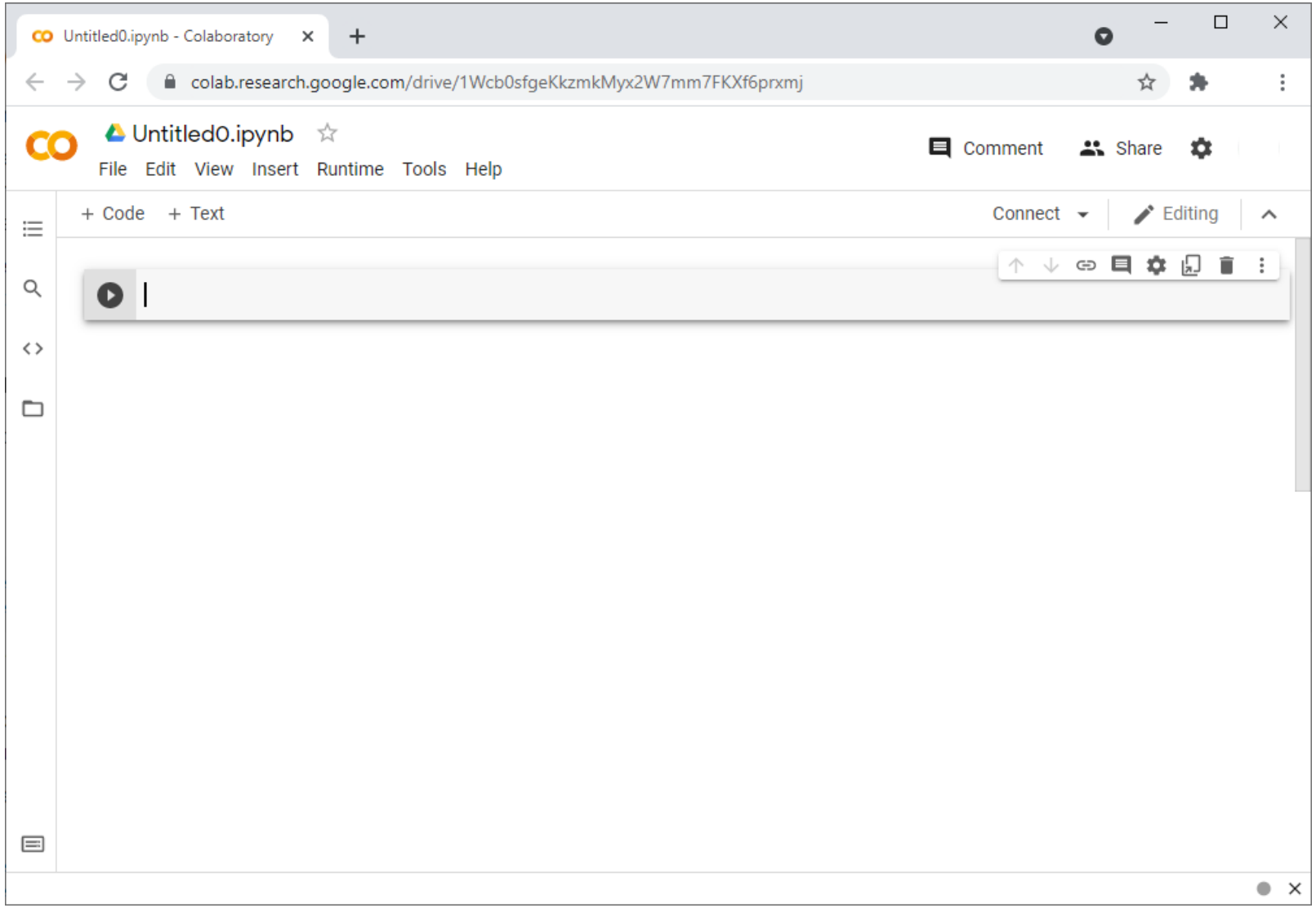
- Type `jupyter lab` and press Enter
  - Jupyter Lab should start up just like it did when you launched it from Anaconda Navigator by via a new browser window



ALTENATIVE: Google Colab or Colaboratory

- Freemium product by Google research, based on Jupyter
- Accessed via browser like Jupyter but connection to Internet needed
- **No need to install anything on your machine (e.g. Anaconda, JupyterLab, etc.) – runs on Google servers**
- Free computation power (GPU and TPU)
- Almost all important libraries are pre-installed
- Colab's notebook files are stored in your google drive, so they can be accessed from anywhere
- Allows for sharing notebooks with other people without even downloading them
- Start using Google colab: https://colab.research.google.com/
- Disadvantages: Limited resources (RAM, CPU) unless you use the paid version, dependency on internet connection.

Google Colab Notebook



Google Colab Notebook